Foundations of Generative AI (ICT4403)

Assignment 2: Autoregressive Text Generation

Framework: PyTorch | Dataset: Tiny Shakespeare

1. Objective

Compare Simple RNN, Bidirectional RNN, LSTM, and GRU models for next-token prediction and autoregressive text generation.

2. Dataset and experimental setup

Tiny Shakespeare source: https://raw.githubusercontent.com/karpathy/char-rnn/master/data/tinyshakespeare/input.txt. The run used lowercase word/punctuation tokenization, a chronological 80/10/10 split, 12,000 tokens, context length 12, embedding size 96, hidden size 128, batch size 128, Adam at 0.002, and two epochs per model.

3. Quantitative results

Model	Test loss ↓	Perplexity ↓	Accuracy ↑	Time (s)
RNN	6.0009	403.80	11.20%	3.77
BIRNN	6.0370	418.64	10.86%	5.15
GRU	6.0441	421.61	11.36%	5.68
LSTM	6.0476	423.08	11.87%	6.95

Lowest test loss: RNN (6.0009); highest accuracy: LSTM (11.87%); fastest training: RNN (3.77 s). The high perplexity and modest accuracy show that this lightweight two-epoch run has limited predictive quality.

4. Temperature analysis

Text was generated from shared seed sequences at temperatures 0.5, 1.0, and 1.3. Lower temperature generally increased repetition; higher temperature generally increased lexical variety and reduced repeated bigrams. However, generated samples did not consistently preserve grammatical coherence or context.

Temperature	Mean unique-token ratio	Mean repeated-bigram ratio
0.5	0.461	0.275
1.0	0.584	0.067
1.3	0.566	0.043

5. Conclusion

For this run, RNN achieved the lowest test loss and fastest training, while LSTM achieved the highest next-token accuracy. No model dominated all metrics. The outputs remain limited, likely due to the small corpus cap and short training duration. More epochs and more training data would be needed for a stronger comparison.

6. Reproducibility

The project outputs include measured metrics, per-epoch histories, generated samples, temperature diagnostics, plots, and trained checkpoints. Add the actual GitHub repository URL before final submission.